

LESSONS FROM EARTHQUAKE RECOVERY FOR FLOOD RECOVERY



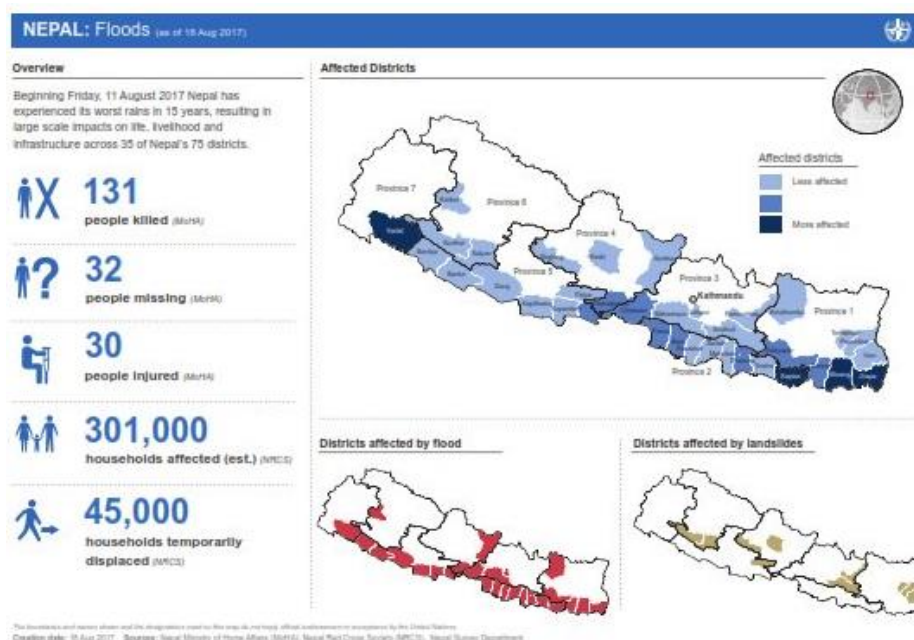
Aerial view of flooding in Dhanusha, 13 August. Photo credit: Niranjana Shrestha.

1. [Leadership and coordination under a single programme](#)
2. [Be realistic about the timeframe required and plan and budget accordingly](#)
3. [Focus on results instead of inputs](#)
4. [Effectiveness of GoN financial grants reliant on link to compliance and presence of comprehensive socio-technical assistance](#)
5. [Coverage, equity, and inclusion](#)
6. [Engage and develop community leaders, elected officials, and construction material vendors and producers as effective agents of the recovery](#)
7. [Land issues and longer-term resilience](#)
8. [Access to finance and indebtedness, impact of remittances and migration](#)
9. [Focus on developing compliance requirements, not specific designs, for traditional as well as emerging trends](#)
10. [Recovery is more than reconstruction of houses](#)

Introduction

This document brings together lessons from HRRP staff, as well as lessons gathered from discussions with the HRRP Strategic Advisory Group (SAG) and other HRRP partners, and from experience from the day of the earthquake, through the Shelter Cluster, up to today, almost three years since the April 2015 earthquake.

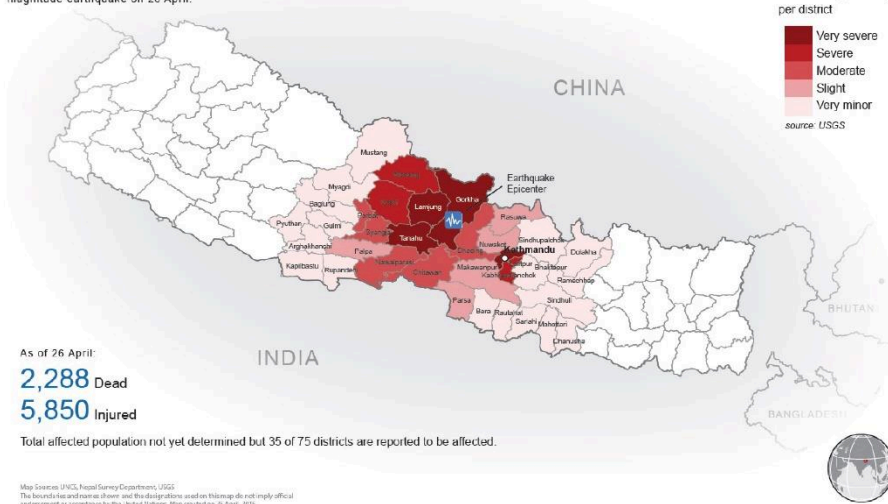
Whilst there are many differences between the April 2015 earthquake and the August 2017 floods, not least in terms of the type, context, scale, and impact of the disaster, we hope that bringing together these lessons from the earthquake recovery will prove useful for those supporting recovery from the floods. When developing the lessons presented below, the similarities between the two recovery efforts have also been considered particularly that housing is the largest sector in both recovery efforts and that the Government of Nepal (GoN) is distributing reconstruction and retrofit grants to affected households.



UN RCO Flood Impacts Infographic, 18 August 2017

NEPAL: Earthquake Humanitarian Snapshot

On 25 April, a 7.9 magnitude earthquake struck Nepal, with the epicenter in Lamjung District (north west) of Kathmandu. Dozens of aftershocks followed, including a 6.7 magnitude earthquake on 26 April.



OCHA April 2015 Earthquake Impact Infographic, 26 April 2015

Having the two recovery efforts running side by side presents a unique opportunity for collaboration and learning, and the flood recovery is ideally positioned to draw on all the positives from the earthquake recovery, and to avoid the challenges that have been faced.

1. Leadership and coordination under a single programme

The early development of a Government-led housing recovery programme, harnessing the contributions and activities of POs under a common identity, with collective and shared goals and responsibilities, agreed standards and approaches encompassing a range of actors and activities would have many benefits. **Without a common programme, there is a risk of fragmentation, uneven levels of assistance, gaps where there is no assistance, confusing and contradictory advice, inefficient individual efforts to develop tools, and limited information on what is being provided, where and to whom – all of which has been seen under the post-earthquake housing reconstruction.**

A common programme, with common agreements (MOU / TOR) for the range of core TA activities to be provided by POs at Rural / Urban Municipality level can help to guide the PO programme planning, provide predictability for Government and communities, and can streamline the administrative approval, monitoring, and reporting processes. Common TORs can also reinforce coverage and balance in the levels of assistance and optimise use Partner Organisation capacities and resources. The collective development of technical assistance resources will optimise the use of expertise and funds, ensuring benefit for all agencies and the wider response.

Developing and maintaining a common approach and collaborative work environment across all the government bodies

This has been one of the major challenges for the post-earthquake housing reconstruction. For example, in the post-earthquake housing reconstruction there have been many issues with the inclusion of household WASH as part of the housing reconstruction programme. WASH and shelter / housing were very separate during the cluster / emergency phase and this separation continued through to the recovery, creating challenges for the NRA and the Department of Water Supply and Sanitation (DWSS) to bring together the two parts of recovery and creating challenges for Partner Organisations (POs) in relation to development and approval of projects for the two sectors. **Having a single programme, rather than a collection of projects, will support coordination and cooperation between the different GoN ministries, departments, and agencies that will be involved in the flood recovery.**

Lack of funding committed to socio-technical assistance by Partner Organisations (POs)

There was clear direction from the Shelter Cluster and GoN early on that the housing reconstruction grant would be provided by the government, and that POs were requested to provide the accompanying socio-technical assistance to support this. In August / September 2015, 61 POs responded to a Shelter Cluster recovery and reconstruction survey reporting that they had [funding of more than 200 million USD already in place specifically for housing recovery and reconstruction](#). Based on Shelter Cluster estimates, this funding from POs could have been used to deliver the minimum TA package across all 31 districts and there would have been a significant pot of money left over. **Unfortunately, 14 POs chose to commit resources to providing the GoN housing reconstruction grant to a total of 17,457 HHs, or 2.8% of the total number of eligible households in the 14 districts – even though this was already covered by the government.** These funds could have been used to provide the full package of TA to over 350 VDCs, or nearly 60% of the affected population in the fourteen districts. The funds could also be used to provide top up grants to over 120,000 households. This lack of vision of the bigger picture, and view of PO projects as independent of the overall

recovery programme, has had a hugely negative impact on the overall recovery. As of 25 December 2017, HRRP 4W data shows that just 4.2% of VDCs in the fourteen most affected districts are receiving the [minimum package of TA support as recommended by HRRP and NRA](#).



Laprak, Gorkha before the April 2015 earthquake. The buildings are traditional stone masonry structures with timber or slate roofs. Photo credit: <http://www.amis-de-laprak.com/laprak-village-01.htm>.



Laprak today – the Non-Resident Nepali association is in the process of directly constructing 100s of houses in the village outside of the GoN housing reconstruction programme. Photo credit: <https://goo.gl/yEB7R3>.

2. Be realistic about the timeframe required and plan and budget accordingly

Focus on the longer term (at least 5 years, but good to acknowledge that recovery will likely take more than 5 years) early and plan accordingly. Budgeting and allocation of resources, especially human resources and associated logistics, needs to align with a realistic timeframe. Owner driven reconstruction will move at the pace of households; don't think you can fully plan or control shelter or housing, you have to continuously respond to the momentum people drive themselves. At best, you can inform, incentivise, guide, assist. In an owner driven reconstruction, we need to trust households and communities to manage the process, providing support as and when required.

Building works do not go through the full year with monsoon and winter impacting on the building season in different ways in different areas. Generally, construction runs from roughly September to April / May each year. In this context consideration needs to be given to the winter and monsoon seasons that households will go through in temporary shelter and what additional support may be required.

Impact of short term recovery programme planning

In the earthquake recovery, many POs had recovery programmes which were only 2 years in duration. Given that the reconstruction was never going to be complete in 2 years this represents a lack of realistic planning and now leaves a situation where the limited coverage of PO support is reducing further just as households come into the first major building season. Much of the support to date has also been focused on households that had started reconstruction their homes – about 32.9% of affected households (as per MoUD CLPIU data as of 22 January 2018). With most of the remaining households likely to start construction during this building season and the next, **the absence of comprehensive socio-technical assistance can be expected to have a negative impact on compliance rates.**

Were the NRA tranche deadlines effective?

In July 2017, the [NRA set deadlines for the disbursement of the GoN housing reconstruction grant](#) as follows:

- Grant agreement signed with local bodies by November 16 2017
- First tranche must be disbursed by January 13, 2018
- Second tranche must be disbursed by April 13, 2018
- Third tranche must be disbursed by July 15, 2018

The NRA felt that setting the deadlines would motivate households to speed up their reconstruction. Having now passed the first two deadlines, it is becoming clear that many households will not meet any of the deadlines, and that many people are constructing very small, one storey, one room, houses to secure the tranche but are also planning to rebuild / repair a house outside of the reconstruction programme, using traditional technology, where they will eventually live. The NRA has applied to cabinet for the deadlines to be extended and it is expected that they will be but it is clear that the pressure from the deadlines has not necessarily resulted in positive reconstruction trends.



Makwanpur, IndraSarowar Rural Municipality-4: this stone and mud mortar masonry house with timber banding and flexible roof has been inspected and approved for the second and third tranche of the GoN housing reconstruction grant. The tentative cost of this building is 150,000 NPRs, excluding the home owner's contribution and the cost of local material used. Such one room, one-storey houses present issues for home owners as they do not meet living requirements for families used to residing in larger, multi-storey houses.

3. Focus on results instead of inputs

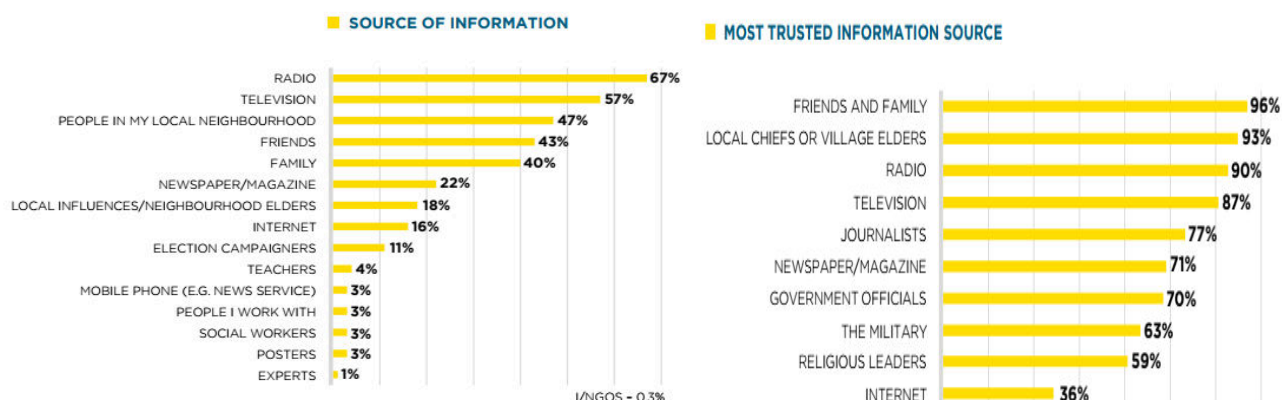
Data collection for the earthquake response has focused heavily on input (and often mostly the input of POs, and to a much lesser extent GoN), when the story of the shelter and housing efforts post-earthquake was what people did themselves with some support from Partner Organisations (POs) and GoN. Having a way to assess the outcomes from the support provided by POs and GoN is important to understand its effectiveness.

For example, the post-earthquake housing recovery and reconstruction has used large-scale cash response for temporary shelter, winterisation, and housing reconstruction (and will now be replicated for the flood response). From the data collected by the Shelter Cluster, HRRP, and GoN we know the amount of cash distributed, where cash was distributed, etc. but what are the outcomes? How is this tracked?

This is equally important for the associated socio-technical assistance which should accompany the cash disbursement. For example, a huge amount of energy, time, and resources was invested in the preparation of a huge number of different leaflets and posters relating to temporary shelter and housing recovery and reconstruction. What was the impact of these? What was the impact of there not being a harmonisation of messaging? In the case of the Shelter Cluster 10 Key Messages posters the content was superseded by

government validated standards but the posters remained in circulation causing confusion, and leading to problems for people who followed this advice.

In 2016, BBC Media Action conducted a nationally representative study in 25 districts, with a total of 4,000 respondents. Respondents were asked about their sources of information on current affairs and political issues, as well as about which of these sources they most trusted. Results demonstrate that radio, television, and family / friends / neighbours are not only major sources of information, but are also considered trusted sources. Social workers, posters, and I/NGOs make up a small portion of people's sources of information, and do not even feature in the trusted sources category.



What has been the impact of conducting training for 50,000+ masons?

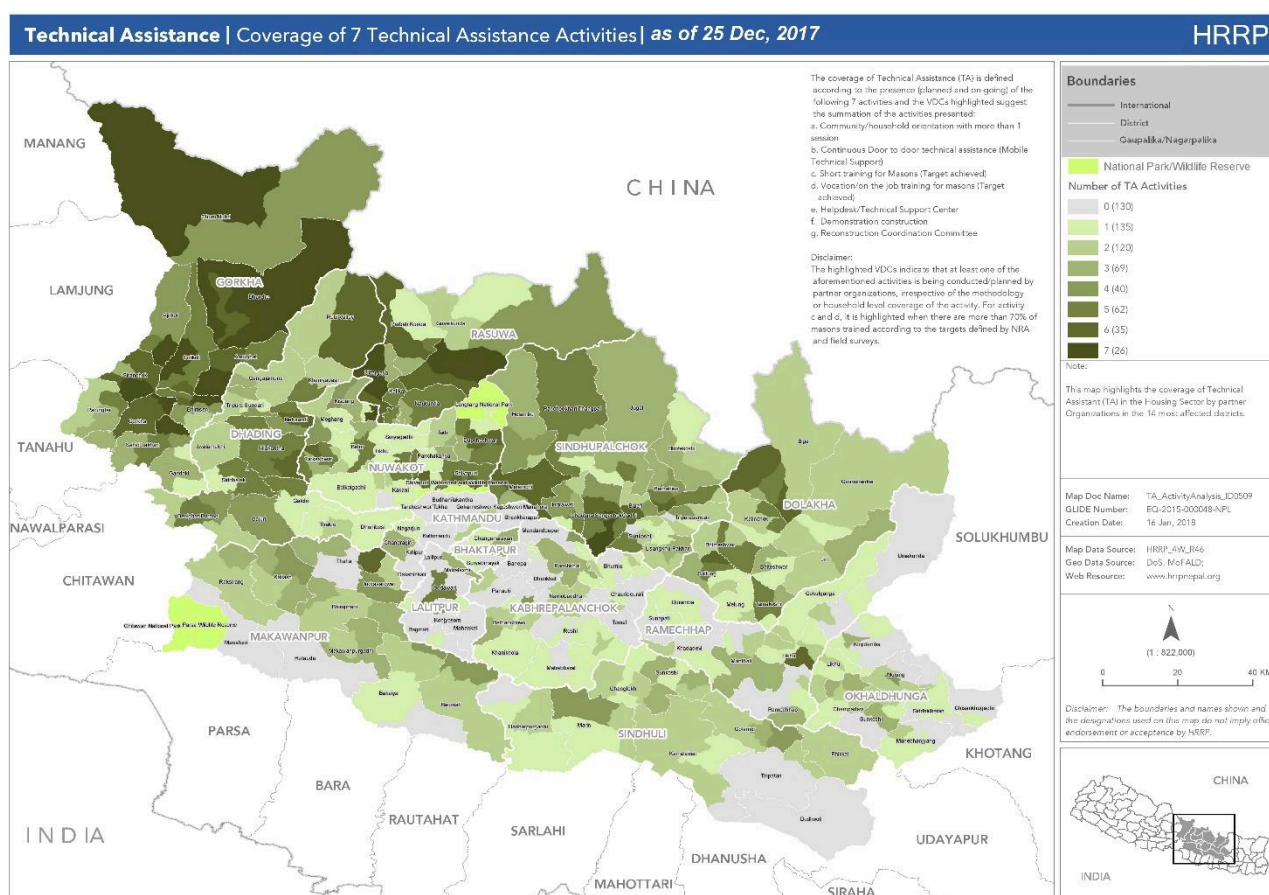
A [JICA survey conducted with trained masons in 2016](#) found that more than 90% of the masons surveyed knew the compliance requirements, but only 22% of the houses they had worked on were compliant. JICA found that there were three main factors causing this; the home owner makes the decisions about the design not the mason, masons require refresher training and continuous support from socio-technical assistance teams to be able to consistently apply safer building techniques, and there needs to be some motivation for masons to remain in their VDC once they have completed the training. Despite the obvious need for additional support for masons beyond training courses, the investment in training courses has not been matched by investment in the required follow up support. For the flood recovery it will be important to learn from this and set up a strong framework for training which goes beyond just the training course itself.

On the other hand, starting as early as possible with demonstration houses to provide visible local launch of Government of Nepal housing recovery programme technical assistance is a valid and useful investment. It can provide an opportunity for households, masons, and community leaders to learn from visual demonstrations of compliant construction techniques.

4. Effectiveness of GoN financial grants reliant on link to compliance and presence of comprehensive socio-technical assistance

The disbursement of the GoN financial grant of 300,000 NPRs for reconstruction is hugely positive and provides an equitable level of support to all households that is tied to, and incentivises, reconstruction that is compliant with building code standards.

However, the positive impact of this financial support can only be maximised through provision of comprehensive socio-technical assistance. The availability of information, advice, and training as early as possible, and programmed over a realistic timeframe, to support people to make informed decisions about reconstruction is critical. Socio-technical assistance also needs to be flexible to scale up and / or adjust to meet households needs as they progress and change over the course of the recovery. Unfortunately in the post-earthquake housing reconstruction coverage of socio-technical assistance by POs has always been poor and as of 25 December 2017, HRRP 4W data shows that just 4.2% of VDCs in the fourteen most affected districts are receiving the [minimum package of TA support as recommended by HRRP and NRA](#):



A key learning from the earthquake response would be to take care with communications and make every effort not to confuse people. **Effective socio-technical assistance requires trained staff who have an established support framework providing them with the ongoing support required to ensure they have correct, and up to date information to provide to households and that they have the skills to convey complex information effectively.** To date in the post-earthquake housing reconstruction there has been an

over emphasis on deploying engineers to the field, with insufficient focus on deploying local masons and social mobilisers. There has also been no investment in developing the communication skills of field staff or establishing a framework for ensuring a harmonised approach amongst field staff so there have been numerous issues with misinformation confusing communities and households and reducing trust in any information provided.

Results from the October 2017 Common Feedback Project reconstruction survey question, ‘what do you know about the grant process?’ provide interesting insight into how low the levels of knowledge on the key parts of the reconstruction programme are amongst affected households. Only 22% of respondents knew that the GoN engineers must inspect and approve the house for the second and third tranche. 28% of respondents thought that they had to reconstruct their home using one of the designs from the DUDBC Design Catalogue Volume 1. Only 33% of respondents knew that they could only receive the second tranche after the foundation had been constructed and approved by a GoN engineer.

What do you know about the grant process?

First Tranche 50,000

**Second Tranche
150,000**

Third Tranche 100,000

50,000 is first
tranche (91%)

150,000 is second
tranche (69%)

100,000 is third
tranche (51%)



Government
approves house
design (33%)



Second tranche
after foundation
(33%)



17 models
(28%)



Engineer inspect
and sign off each
tranche (22%)

5. Coverage, equity, and inclusion

The earthquake is estimated to have pushed an additional 750,000-900,000 Nepalese living close to the poverty line into poverty. It is likely that the flooding has similarly pushed those living close to the poverty line further into poverty; especially considering the floods are estimated to have caused nearly 70 million USD of damage in the agriculture sector and almost 103 million USD of damage to livestock ([Nepal Flood 2017, Post Flood Recovery Needs Assessment](#)). **Reconstruction actors will need to make sure that recovery efforts are inclusive and that appropriate, targeted support is available to marginalised and vulnerable households to support with their overall recovery.**

Recovery actors should also think and plan in terms of scale, coverage, and replicability to add value to the very large number of households undertaking reconstruction, and over a very wide area. In the earthquake response there have been challenges with POs providing reconstruction support in 4 out of 9 wards in a VDC! POs should stretch support provided to cover as large an area as possible, rather than getting into micro coverage projects.

Inclusive socio-technical assistance must consider gender, language, ethnicity, and literacy levels of the households

In the post-earthquake housing reconstruction **women have consistently reported not having been meaningfully engaged in the reconstruction process, or treated as agents of the recovery.** This has played out in various ways; from the beginning of the enrolment process when women's names were not listed for bank accounts making them unable to access the reconstruction grant, to a persistent gap in targeting them with reconstruction information in a way that meets their communication needs. The Common Feedback Project (CFP) October 2017 survey included the question “do you feel that men and women are equally engaged in the reconstruction process?”. 60% of male respondents felt they were equally engaged, as compared to 49% of female respondents and some of the most common responses to account for women's unequal engagement in the reconstruction process, from both male and female respondents, included: “women are not as strong as men” and “women don’t know how to reconstruct”. If more than 50% of the population are not encouraged and supported to take part in the reconstruction it is unlikely that the expected results can be achieved.

Other factors of vulnerability / marginalisation such as ethnicity and disability have also impacted greatly on post-earthquake housing reconstruction. Round four of The Asia Foundation ‘Independent Impacts and Recovery Monitoring’ Project, conducted in April 2017, found that 70% of respondents felt they had been able to prepare their shelters for winter weather, however, 17% failed to repair their shelters completely and 11% failed to repair them sufficiently for winter. **Those who were unable to prepare their shelter for winter were more likely to be from low caste groups.** The report also indicates that those with lower income were less likely to be able to prepare their shelter for winter. People with disabilities also faced greater difficulty repairing their shelters, with only 46% of people with disabilities able to carry out repairs.

The **Common Feedback Project Flood Perception Survey, October 2017 identified a worrying trend with 34% of respondents saying they did not feel respected by aid providers, including government and I/NGOs.** 44% of respondents said it was because there was not enough assistance available for all entitled to it, and 43% said the assistance was not fairly or evenly distributed. Hopefully the long term flood recovery can reverse this trend through an inclusive, respectful, and equitable programme of support.

6. Engage and develop community leaders, elected officials, and construction material vendors and producers as effective agents of the recovery

Since the 2015 earthquake, Nepal has gone through enormous change in relation to governance structures. The first elected local officials in more than 20 years are now in place, and they have a key role to play in the recovery. Where the earthquake recovery has had to adjust and adapt to these new structures, the flood recovery can immediately be established to work with and support local elected officials so that they can effectively participate in the recovery process.



A woman casts her vote in the first round of the local elections in May 2017. Photo credit: <https://goo.gl/doyc2k>.

Community leaders and local elected officials are often the main source of information on the recovery for people in affected areas. They need support to share information effectively and to make sure that they can provide timely and correct information (and to avoid rumours and misinformation). This is especially critical in the flood affected areas where already politicisation of the response has been reported through the Common Feedback Project (CFP) Flood Perception Survey from October 2017; 6% of respondents reported having been asked to exchange something for relief or support (and a further 14% refused to answer). Among those who said they were asked to exchange something, 94% were asked to give political support or a vote. In particular, **elected officials who are female, and / or from marginalised groups, need to be supported to take on the role of effective agents of recovery so that they can bring other women and members of marginalised groups into the process.**

Construction material vendors and producers are the first point of contact for households regarding material purchase and planning and need to be able to provide accurate information on material quality and the options available to households. They need to be a reliable source of information on compliance requirements so that they can support households in decision making and purchasing materials.

7. Land issues and longer-term resilience

Housing and settlement recovery must consider land issues, including land at risk of future hazards, land use planning and development control, and property status. **The geo-hazard survey conducted by NRA in the earthquake affected districts, and the subsequent policy development to provide 200,000 NPRs for purchase of land, represents a potential basis upon which to use assessment of risk and hazard to determine eligibility and need for mitigation and / or relocation options.** As the support for landless households is just starting to roll out it remains to be seen how effective it can be, but a similar approach may be useful to apply in flood areas.

In terms of property status adequate consideration needs to be given to social and legal issues which may impact on reconstruction, e.g. renters, land title, landless. The most vulnerable households may remain outside of the reconstruction programme because of land issues related to existing vulnerability / marginalisation. Developing suitable policies and programmes to support these households will be critical to ensuring the success of the reconstruction and making sure that no families are left behind.



Recovery planning must also consider the annual flood risk and the potential for this to impact on recovery on an annual basis. Linking with and incorporating Disaster Risk Reduction (DRR) activities into recovery activities will be essential to ensuring that recovery provides flood affected households with longer term resilience. For example, the District Disaster Relief Committees (DDRCs), and the District Lead Support Agencies (DLSAs), have seen significant impact in flood areas around annual response planning and mitigation strategies. Building these into the recovery from an early stage will strengthen the long-term impact of recovery efforts and reinforce the ongoing DRR efforts in flood affected areas.

Another example is the set of [9 minimum characteristics](#) (9MCs) of disaster-resilient communities developed by Flagship 4 of the Nepal Risk Reduction Consortium (NRRC). These indicators were designed in consultation with GoN, I/NGOs, UN, donors and Red Cross.

The minimum characteristics are aimed at ensuring communities receive consistent community based disaster risk reduction support as well as providing a means to track progress towards increasing disaster resiliency at VDC level across the country. [A review of the 9 MCs](#) provides some detailed guidance on how to incorporate DRR into development planning. Utilising such existing DRR approaches would certainly strengthen the flood recovery efforts.

8. Access to finance and indebtedness, impact of remittances and migration

The financial assistance from the GoN is not intended to cover the full cost of reconstruction and so households need to put their own funds into their reconstruction. In earthquake areas this has been through either savings and / or loans. **Many people taking loans access them through informal channels at extremely high interest rates (as high as 43%).** This also relates to the speed at which the reconstruction grant can be disbursed; if it arrives with houses relatively quickly there is potential for less borrowing and many households have reported borrowing with the intention to repay once they receive the tranches of the GoN housing reconstruction grant. With higher levels of vulnerability and marginalisation, land issues, etc. in the flood affected areas access to finance may be more challenging and households may require extra / innovative support to access financial services that meet their needs.

The impact of remittances and migration on the post-earthquake recovery are not well understood but anecdotal evidence does indicate that it has an impact. **With high rates of migration in flood affected areas before the August 2017 floods, it can be expected that this will play a role in the flood recovery and it is worth investing in research to better understand the impacts and how the positive impacts can be maximised and negative impacts can be minimised.**

9. Focus on developing compliance requirements, not specific designs, for traditional as well as emerging trends

Local building traditions can represent rich sources of knowledge including hazard resistance. Understanding how people live, their cultural and social values and preferences, is critical to understanding how they will build their homes. **Analysing housing needs is more than an engineering exercise.** The compliance menu, and associated inspection formats, should accommodate traditional typologies but must also make provisions for new building materials and typologies, e.g. hollow concrete blocks. They must also consider people's living requirements, e.g. storage space (attics), space for livestock / business, number of storeys, etc.

For the post-earthquake housing reconstruction, the idea that households must construct using one of the designs from the [DUDBC Design Catalogue Volume 1](#) has proved to be one of the most persistent pieces of misinformation. The concept of preparing the design catalogue was to provide examples of a variety of options for housing reconstruction in terms of cost, size, layout, and typology. It is not, and never has been, mandatory for households to select a design from this catalogue, and they are free to prepare house designs outside of the catalogue but these designs must comply with the National Building Code. However, it remains the case that many socio-technical field staff (GoN and POs) and households believe that these designs must be followed to be eligible for the GoN housing reconstruction grant.

The [DUDBC Design Catalogue Volume 2](#), which includes designs for a number of alternative or emerging construction typologies further highlights that the focus should be on compliance requirements rather than specific designs. Many of the typologies, including Compressed Stabilised Earth Blocks (CSEB), for which designs are included in the catalogue are still not included in the inspection process as the relevant compliance requirements and inspection forms have yet to be finalised.

The [Post Flood Recovery Needs Assessment](#) states that the “government assigned NRA and DUDBC to come up with appropriate housing designs in six months”. Based on the lessons from the post-earthquake housing reconstruction it would be advisable to consider adjusting this approach and focusing on developing the compliance requirements associated with disbursement of GoN financial assistance rather than focusing on developing specific house designs.

10. Recovery is more than reconstruction of houses

Shelter and settlement recovery involves processes of physical, social, and economic recovery. This means more than reconstruction of housing, and more than individual family level activities. Recovery needs to include community and settlement level activities, water and sanitation, community and social infrastructure and services, livelihoods, supply chains, communication and information processes, mobilisation, and institutional arrangements.

Housing recovery activities will require integrated and coordinated planning and implementation with other sectors, at community level and at policy and programming level. This requires strong coordination and collaboration across multiple GoN departments and ministries, as well as a wide range of Partner Organisations (POs) – see point 1 above ‘Leadership and coordination under a single programme’.

For the flood recovery this is particularly important as flooding is beyond individual households to mitigate and it should be expected that watershed management and / or infrastructure works will be required.

The village of [Tilathi in Saptari District](#) provides an excellent reminder of this. When more than 150 houses in the village were flooded during the August 2017 rains, it was the fourth time in two months that flooding had affected the village. Villagers lost all their household goods, food grains, livestock, and their homes. As the villagers began the hard task of recovering yet again from floods, they refused emergency relief from the Government and NGOs. The villagers instead want a permanent solution to the flooding which they feel is a permanent dyke to protect the village from the Khado and Jitta rivers.

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